Example 6.4

If you toss a fair coin three times, the outcome of an experiment consider as random variable which counts the number of heads on the upturned faces. Find out the probability mass function and check the properties of the probability mass function.

Solution:

Let X is the random variable which counts the number of heads on the upturned faces. The outcomes are stated below

Outcomes	Values of X
(HHH)	3
(HHT)	2
(HTH)	2
(THH)	2

Outcomes	Values of X
(HTT)	1
(THT)	1
(TTH)	1
(TTT)	0

These values are summarized in the following probability table.

Table 6.3

Value of X	0	1	2	3	Total
$p(x_i)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	$\sum_{i=0}^{3} p(x_i)=1$

(i) $p(x_i) \geq 0 \ \forall \ i$ and

(ii) $\sum_{i=0}^{3} p(x_i)=1$

Hence, $p(x_i)$ is a probability mass function.

Example 6.5

Two unbiased dice are thrown simultaneously and sum of the upturned faces considered as random variable. Construct a probability mass function.



Fig.6.2

Solution:

Sample space $(S)=\begin{Bmatrix} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{Bmatrix}$

Total outcomes : $n(S) = 36$

Table 6.4

Out comes	(1,1)	(1,2) (2,1)	(1,3) (2,2) (3,1)	(1,4) (2,3) (3,2) (4,1)	(1,5) (2,4) (3,3) (4,2) (5,1)	(1,6) (2,5) (3,4) (4,3) (5,2) (6,1)	(2,6) (3,5) (4,4) (5,3) (6,2)	(3,6) (4,5) (5,4) (6,3)	(4,6) (5,5) (6,4)	(5,6) (6,5)	(6,6)
Sum of the upturned faces (X)	2	3	4	5	6	7	8	9	10	11	12
$P_X(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Discrete distribution function

Definition 6.4

The discrete cumulative distribution function or distribution function of a real valued discrete random variable X takes the countable number of points $x_1, x_2, \ldots$ with corresponding probabilities $p(x_1), p(x_2), \ldots$ and then the cumulative distribution function is defined by

$$F_X(x)=P(X\leq x), \text{ for all } x\in R$$

$$i.e., F_X(x)=\sum_{x_i \leq x} p(x_i)$$

For instance, suppose we have a family of two children. The sample space

S = {bb, bg, gb, gg}, where b =boy and g = girl

Let X be the random variable which counts the number of boys. Then, the values (X) corresponding to the sample space are 2, 1, 1, and 0.

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